

Lab 5 - Newton Method

1. Introduction

In this laboratory work, Newton's method was studied for both one-dimensional and multidimensional problems. The dependence of the method on the initial guess was investigated using the function $f(x) = x \sin(4x)$. The Newton method for systems of equations and Aitken acceleration were also implemented.

2. Newton Method for Multiple Roots

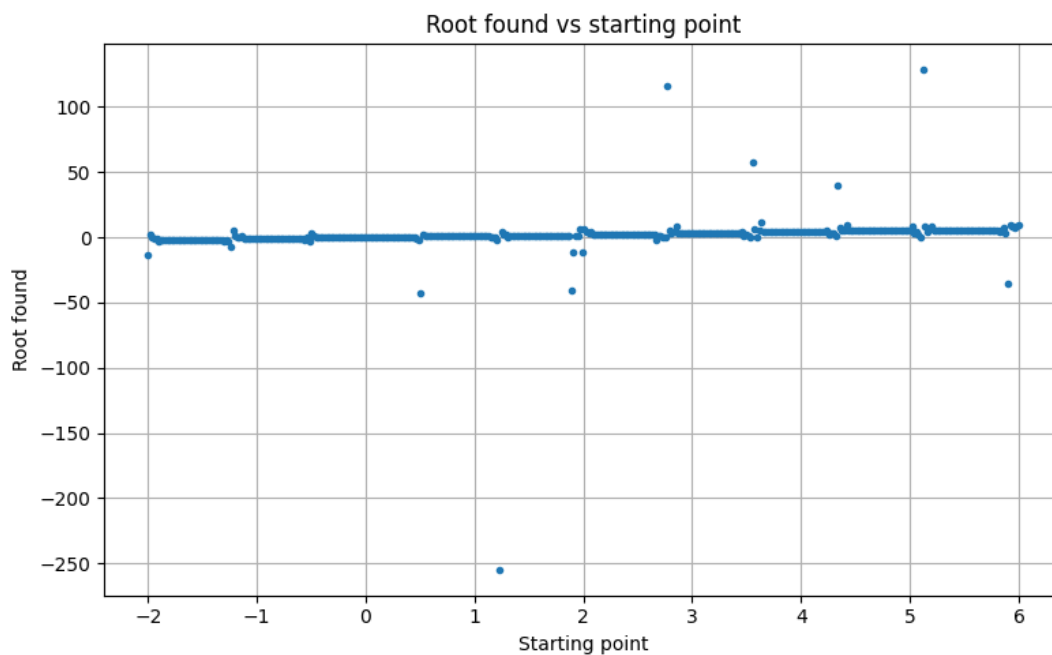
2.1 Function Definition

The following function was analyzed:

$$f(x) = x \sin(4x)$$

This function has multiple roots because:

- $x = 0$ is a root



$$\sin(4x) = 0 \text{ when } 4x = k\pi$$

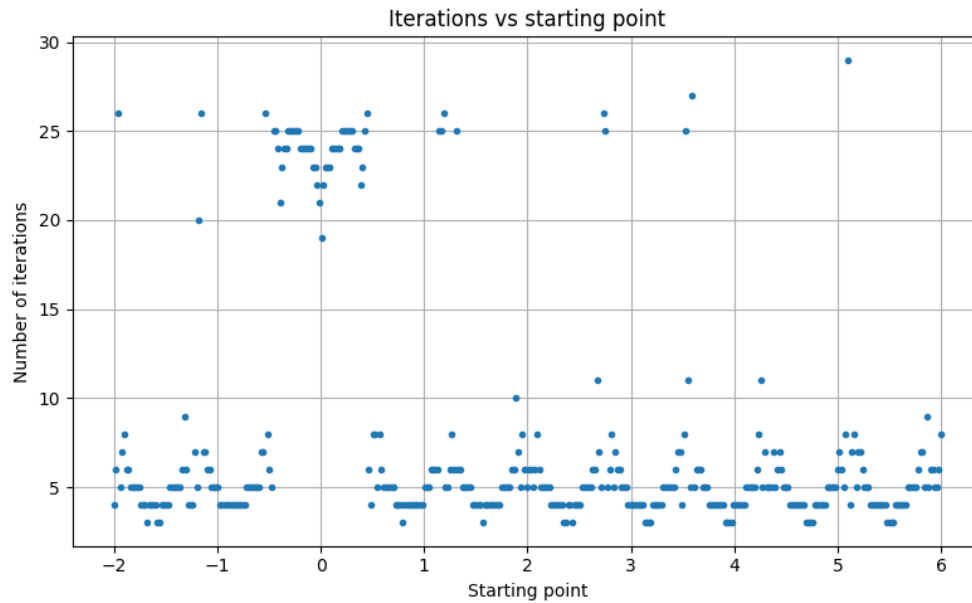
- Therefore $x = k\pi / 4$

2.2 Root Found vs Starting Point

Newton's method was applied for many different starting points.

The graph shows that different initial guesses converge to different roots.

The relationship is not continuous because small changes in the starting point may lead to completely different roots.



2.3 Number of Iterations vs Starting Point

The number of iterations depends strongly on the starting point.

For most values, convergence occurs in a small number of iterations.

However, in some regions the number of iterations increases because the derivative becomes close to zero.

3. Newton Method for Systems of Equations

3.1 System of Equations

$$x^2 + y^2 + z^2 = 2$$

$$x^2 + y^2 = 1$$

$$x^2 = y$$

3.2 Function Vector and Jacobian

Function vector:

$$F1 = x^2 + y^2 + z^2 - 2$$

$$F2 = x^2 + y^2 - 1$$

$$F3 = x^2 - y$$

Jacobian matrix:

$$J = \begin{bmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \\ 2x & 1 & 0 \end{bmatrix}$$

[2x, 2y, 0],
[2x, -1, 0]]

3.3 Numerical Solution

The Newton method converged to the following solution:

$x \approx 0.78615138$
 $y \approx 0.61803399$
 $z \approx 1.00000000$

The solution was obtained after 4 iterations.

The check vector was approximately zero, which confirms the correctness of the result.

Solution:

[0.78615138 0.61803399 1.]

Iterations:4

Check:

[0.00000000e+00 -1.11022302e-16 0.00000000e+00]

4. Aitken Acceleration

4.1 Newton Sequence

The Newton sequence converged to approximately:

$x = -5.49778714$

Newton sequence:

[0.5 -5.40424092 -5.50422863 -5.49779325 -5.49778714 -5.49778714
-5.49778714 -5.49778714 -5.49778714 -5.49778714 -5.49778714]

Aitken accelerated sequence:

[-5.50595108 -5.4981824 -5.49778714 -5.49778714 -5.49778714 nan
nan nan nan]

4.2 Accelerated Sequence

Aitken acceleration improves convergence speed by using three consecutive approximations.

The accelerated sequence converges faster than the standard Newton sequence.

After convergence, NaN values appear because the denominator becomes very close to zero.

5. Discussion

The results show that Newton's method strongly depends on the initial guess.

Different initial guesses may converge to different roots.

The multidimensional Newton method successfully solved the nonlinear system.

Aitken acceleration improved convergence speed in the one-dimensional case.

6. Conclusion

Newton's method is a powerful numerical method for solving nonlinear equations.

However, the convergence depends on the initial guess and the behavior of the derivative.

For multidimensional systems, the Jacobian matrix is essential.

Aitken acceleration can improve convergence speed, although numerical instability may appear after convergence.